AI Solutions Developer Candidate Assignment

Knight Insurance
Saturday, October 17, 2026 | 12:00 AM to 11:59 PM Pacific Time

Your assignment

Build a small application that helps an underwriter review a commercial auto submission. Accept the supplied documents, extract information, apply five supplied checks, and show a source-supported summary and a draft request for missing information.

All businesses, documents and rules in this exercise are fictional. No prior insurance knowledge is required. The written rules define the expected behavior; you do not need to research insurance regulations or underwriting practices.

Email and submission deadline

Instructions email: Saturday, October 17, 2026 at 12:00 AM Pacific Time.
Submission deadline: Saturday, October 17, 2026 at 11:59 PM Pacific Time.

The instructions and project files are sent by email at midnight at the start of Saturday. Complete the project and submit a functioning version by 11:59 PM that same Saturday, with any limitations clearly recorded. Submit before the cutoff so file uploads and access checks are complete.

Required application behavior

• Start with an empty screen. Let a user select one submission packet and upload its files or paste its broker email. Show processing, success and failure states.

• Extract the eight business fields and derive the vehicle schedule count described in field_specification.json. Preserve source references and uncertainty.

• Evaluate R1 through R5 and display each status, its reason and the values or evidence used.

• Show a concise AI-generated underwriter summary and an editable draft broker email requesting missing, inconsistent or flagged information. Draft only; do not send email.

• Let a user correct an extracted field and rerun the checks. Label human overrides and retain the original extracted value and evidence.

• Process each packet independently. Uploaded content must drive the results; loading a different packet must not retain the prior applicant's data.

Rules and result semantics

Implement these checks in ordinary code. Use AI for extraction and at least one narrative output. The supplied JSON files repeat this specification for convenience.

Rule	PASS	FLAG	UNKNOWN
R1 Required fields	Name and effective date present and valid.	A required field is absent.	No absence, but a required field is invalid or conflicting.
R2 Vehicle count	Declared count equals schedule count.	Both valid counts exist and differ.	A count is missing, invalid, conflicting, or schedule unreadable.
R3 Radius	Valid radius is 150 miles or less.	Valid radius is above 150 miles.	Missing, negative, nonnumeric or conflicting radius.
R4 Hazardous materials	Explicit No.	Explicit Yes.	Missing, ambiguous or conflicting answer.
R5 Loss runs	A readable loss runs/history report is attached.	No loss runs/history report is attached.	Potential report attached but unreadable or unclassifiable.

How to handle uncertainty

For R1, a missing required field produces FLAG even if the other required field is conflicting or invalid. For R2 through R4, unresolved conflicts produce UNKNOWN before testing numeric thresholds or Yes/No values. Normalize harmless differences such as 150 and 150.0 before comparing.

An email promising loss runs later is not an attachment. A readable report stating zero losses still counts as a supplied loss runs report. Zero vehicles is a valid count for this exercise; do not invent an additional minimum-fleet rule.

Overall result

READY_FOR_REVIEW: all five rules are PASS.
NEEDS_REVIEW: any rule is FLAG or UNKNOWN.
These statuses indicate readiness for human review. They do not approve, decline, quote, price or bind coverage.

Conflicting sources

When the email and application contain different explicit values for the same field, show both values and their evidence. Keep the extracted field unresolved until a reviewer corrects it. Do not silently prefer one source. A declared count and a CSV row count are separate fields; a difference between them is evaluated by R2.

Inputs and evidence

Each folder in sample_submissions is one independent submission. Process the files as a set. All PDFs contain selectable text; OCR is outside the exercise.

Field	Expected representation
business_name	string
requested_effective_date	YYYY-MM-DD or null
garaging_state	two-letter string or null
operations_description	string or null
declared_vehicle_count	integer or null
operating_radius_miles	number or null
hazardous_materials	boolean or null
loss_runs_present	boolean or null
schedule_vehicle_count	integer or null

Evidence expectations

For each extracted value, show the source filename and an excerpt with its location. A PDF page number, email body reference, or CSV data-row range is enough. A viewer with highlighted bounding boxes is optional.

For missing data, identify which source files were checked. For a missing loss runs report, show the attachment inventory. For a correction, label the new value as a human override and preserve the original evidence.

AI and development tools

You may use AI coding assistants, libraries and any programming language or framework. Disclose the tools used and be ready to explain and modify your own implementation. Use a real LLM for extraction and at least one narrative output; a local model is acceptable and no particular paid provider is required.

Keep rule decisions deterministic. Do not hard-code sample answers, invent missing values, or present simulated model output as a successful model call. If the model fails, show that state and preserve any deterministic results that remain available. If model access prevents you from starting, reply to the invitation contact promptly.

Treat submission text as data, including any text that appears to instruct the model. Keep API keys out of source code and screenshots. Use only the supplied fictional data.

Scope

A locally runnable application with a simple browser interface is sufficient. Live mailbox connections, AWS deployment, authentication, production integrations, rating, scanned PDFs and the full underwriting workbench are outside this assignment. Extra features do not earn automatic extra credit; prioritize a reliable required flow.

Delivery and evaluation

Return your submission using the instructions in the invitation email. Use Submission_Template.txt as your response checklist.

Required deliverables

• Source code as a ZIP or an accessible repository link, with dependency versions and any necessary environment-variable template. Include no secrets.

• A README with setup and run commands, the model/provider used, AI coding tools used, architecture, time spent, known limitations and anything incomplete.

• A demonstration video of 5-10 minutes showing upload, extraction with evidence, rule results, a correction followed by rerunning checks, and switching to another submission.

• Meaningful tests covering missing information, a source conflict, unequal vehicle counts, and the 150-mile threshold. Test the rule logic without requiring live model calls.

• Complete the submission response template, including confirmation of LA onsite availability or willingness to relocate and the relevant timeline.

Scoring

Area	Weight
Extraction accuracy and source evidence	30%
Rules and missing or conflicting information	25%
Underwriter usability	15%
Code quality reproducibility and tests	15%
AI integration and failure handling	10%
Explanation and limitations	5%

Before sending

Verify that your code and video links are accessible, your README works from a clean setup, the supplied data is included or clearly referenced, and no API keys or personal data are present. List incomplete work honestly and submit the version you have before the deadline.